

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level



MARKING SCHEME

NOVEMBER 2014

CHEMISTRY 9189/5

(a) Award: 1 mark for all burette readings recorded to at least 2 dp (volume of water)

2dp 1 mark for filling in some values of CO_2 volume in the table | Respect of any 2

✓ 1 mark for all correct subtractions in calculating the volume of CO_2

S Up to 3 marks for being able to carry out the experiment as follows:

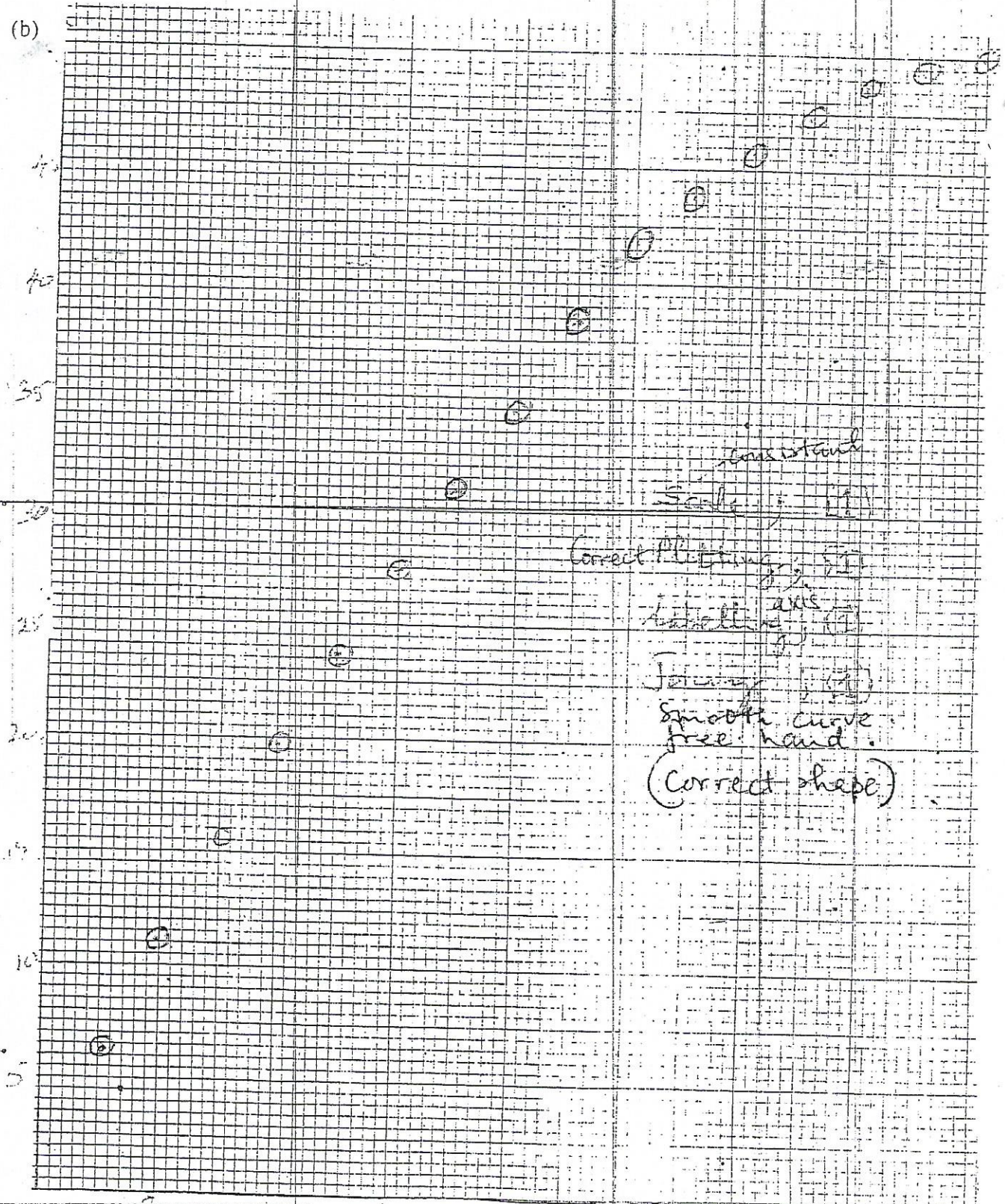
R - 3 marks for recording at least 3 burette readings in descending order

- 2 marks for recording at least 4 burette readings in descending order

- 1 mark for recording at least 5 burette readings in descending order

[Total: 6]

(b)



- (ii) 1 mark for marking the steepest point on the graph
 (1 mark for estimating the correct corresponding time on the graph)
 $\text{gradient} = \frac{\text{change in } y}{\text{change in } x}$ [1]

read candidate's graph;

change in x

- calculate gradient at time in (ii); = Answer [1]
 (units correct) cm^3/min

- (c) (i) - Some $\text{CO}_2(\text{g})$ is dissolved in the water;
 - H_2O vapour is produced which may escape increase volume of gas produced/
 - Air displaced by some acid added.
 - gas may escape if all joints are not air tight/ Any three. [1]

- (ii) 1. tap has a lot of dissolved gases so would dissolve less amount of CO_2 than distilled water;

2. ↑ the surface area for the rxn.
 (iii) increase visibility of burette readings; [1]

- (d) (i) $\text{CuCO}_{3(\text{s})} + 2\text{HNO}_{3(\text{aq})} \rightarrow \text{Cu}(\text{NO}_3)_{2(\text{aq})} + \text{H}_2\text{O} + \text{CO}_{2(\text{g})}$
 ignore state symbols but reject if they are wrong.

- (ii) 1. Correct value of maximum CO_2 obtained from experiment; ECF allowed. [1]

2. maximum value of CO_2 cm^3 moles of CuCO_3 reacted ✓
 24 000 (at r.t.p.)

2dp in standard form

$\text{XM}(\text{CuCO}_3) = \text{mass};$

Repeating the expt several times using lower conc of acid / the vol of tap was

2. (a) 1. Place a methane gas burner filled with methane gas on a balance and record its mass. specify how to measure the mass [1]

2. (Find the mass of an empty 250 cm^3 beaker using a balance.)

3. Pour water in the 250 cm^3 beaker and find mass of the beaker with water. [1]
 (Accept 80-240 cm^3)

4. Insert a thermometer in the water in the beaker and record the temperature of the water. (Place the beaker on a burner stand over the gas burner. (initial) [1]

5. Place a draught shield around the set-up and light the methane gas burner. [1]

6. After a reasonable rise in temperature put off the flame and immediately record the (final) temperature of the water. (temp rise) [1]

7. Place the methane gas burner on a balance and record its mass. [1]